Presentation Layer (Vaadin / MVP)

«UI Shell»
MainLayout
AppLayout + nav

«Session State»
CurrentUser
logged-in member / roles

Views are routed via @Route and rendered inside
Each Presenter depends on Application Services (listed in box) + CurrentUser.

«View»	«Presenter»	Dependencies
LoginView	**LoginPresenter**	MemberService, AdminService, RoleService, ActiveOrderService, CurrentUser
RegisterView	**RegisterPresenter**	MemberService
ProfileView	**ProfilePresenter**	MemberService, CurrentUser
EventBrowseView	**EventBrowsePresenter**	EventService, ProductionCompanyService
EventDetailView	**EventDetailPresenter**	EventService, ShowService, CurrentUser
PurchaseView	**PurchasePresenter**	ActiveOrderService, OrderService, EventService, ShowService, CurrentUser
OrderHistoryView	**OrderHistoryPresenter**	MemberService, EventService, ShowService, CurrentUser
EventManagementView	**EventManagementPresenter**	EventService, ShowService, CurrentUser
CompanyDashboardView	**CompanyDashboardPresenter**	ProductionCompanyService, EventService, RoleService, CurrentUser
RoleManagementView	**RoleManagementPresenter**	RoleService, ProductionCompanyService, CurrentUser
SalesReportView	**SalesReportPresenter**	ProductionCompanyService, CurrentUser
AdminDashboardView	**AdminPresenter**	AdminService, ProductionCompanyService, CurrentUser
VirtualQueuePresenter (within EventDetail/Purchase)		VirtualQueueService, CurrentUser
	LogoutPresenter (invoked from MainLayout)	MemberService, CurrentUser

Routed Views (no presenter)

- «View» **HomeView**
- «View» **AccessDeniedView**
- «View» **SuspendedAccountView**
- «View» **HallLayoutView**
- «View» **AdminPurchaseHistoryView**
- «View» **CompanyManagementView**

External System Interfaces

- « External» *Notifier (Vaadin Push)*
- « External» *Auth*
- « External» *Payment Gateway*
- « External» *Ticket Service*
- « External» *Password Encoder*

Application Layer

- «Interface» *IAuth*
- «Service» **RoleService** — MemberRepo, ProductionCompanyRepo, NotificationRepo
- «Service» **Admin Service** — MemberRepo, ProductionCompanyRepo, NotificationRepo, EventRepo, ActiveOrderRepo, OrderRepo
- «Service» **ActiveOrder Service** — MemberRepo, ShowRepo, EventRepo, TicketRepo
- «Service» **ProductionCompanyService** — MemberRepo, NotificationRepo, EventRepo
- «Service» **EventService**
- «Service» **OrderService** — ActiveOrderRepo, MemberRepo, ShowRepo, TicketRepo
- «Interface» *IPaymentGateway*
- «Interface» *ITicketIssuanceService*
- «Interface» *IPassword Encoder*
- «Interface» *INotifier*
- «Factory» **DiscountPolicyFactory**
- «Factory» **PurchasePolicyFactory**
- «Service» **VirtualQueue Service**
- «Service» **MemberService** — OrderRepo
- «Service» **NotificationService**
- «Service» **Show Service**
- «Service» **Ticket Service**

1

Domain Layer

Notification Aggregate

- «Interface» *INotificationRepository*
- «Factory» **NotificationFactory**
- «Aggregate Root» **Notification**

VirtualQueue Aggregate

- «Interface» *IVirtualQueueRepository*
- «Aggregate Root» **VirtualQueue**
  - List<**Member ID**>
  - **Event ID**

Admin Aggregate

- «Interface» *IAdminRepository*
- «Aggregate Root» ***Admin***

Member Aggregate

- «Interface» *IMemberRepository*
- «Value Object» **Suspension** — * — 1 — «Aggregate Root» **Member**
  - **ActiveOrder ID**
  - List<**Notification ID**>
  - List<**Order ID**>

ProductionCompany Aggregate

- «Interface» *IProductionCompanyRepository*
- «Aggregate Root» **ProductionCompany** — * — 1 — «Value Object» **OrderPolicy**
- «Aggregate Root» **ProductionCompany** — * — 1 — «Value Object» **DiscountPolicy**
- «Aggregate Root» **ProductionCompany** — 1 — 1..* — «Abstract» **Role-State**
  - **Member ID**
- *Manager-Role* — 1 — *Owner-Role* — 1 — *Founder-Role*
- *Manager-Role* — * — 1 — «Value Object» **Permissions**

ActiveOrder Aggregate

- «Interface» *IActiveOrderRepository*
- «Aggregate Root» **ActiveOrder**
  - List<**Ticket ID**>

Order Aggregate

- «Interface» *IOrderRepository*
- «Aggregate Root» **Order** — 1 — 1 — «Value Object» **Customer Details**
  - **Active Order ID**

Ticket Aggregate

- «Interface» *ITicketRepository*
- «Aggregate Root» **Ticket**
  - **Show ID**
  - **Zone**
  - **Seat**

Show Aggregate

- «Interface» *IShowRepository*
- «Aggregate Root» **Show**
  - **Event ID**
- 1 — 1..* — «Entity» VenueMap — 1 — * — «Entity» Zone — 1 — * — «Value Object» **Seat**

Event Aggregate

- «Interface» *IEventRepository*
- «Value Object» **EventPolicy** — 0..1 — * — «Aggregate Root» **Event** — * — 1 — «Value Object» **EventType**
- «Value Object» **DiscountPolicy** — 0..1 — * — «Aggregate Root» **Event** — 1 — 0..1 — «Entity» **Raffle**
  - List<**Member ID**>
- «Aggregate Root» **Event**
  - **ProductionCompany ID**

Legend

- Aggregate Root
- Entity
- Value Object
- Application Service
- Interface (Repository / External)
- Factory
- Aggregate boundary
- View (Presentation)
- Presenter / UI infrastructure